

Framework of the Artificial Intelligence–Inquiry by Engineering Design (AI-IBED)

1. Overview of the AI-IBED Framework

My proposed endeavor is to develop, implement, evaluate, and disseminate an Artificial Intelligence–Inquiry by Engineering Design (AI-IBED) instructional model to strengthen STEM teaching and learning in K–12 schools across the United States. The model will integrate artificial intelligence into an established inquiry-based and engineering-design instructional framework to support teachers in developing scientifically rigorous, engaging, and accessible learning experiences. The endeavor responds to two national needs discussed in Section 7: declining K–12 science achievement and student engagement in scientific inquiry, and the federal priority of preparing U.S. students and teachers to use AI responsibly.

The AI-IBED model builds on the Inquiry by Engineering Design (IBED) framework developed by Brand, Kasarda, and Williams (2017). Their framework combines the 5E instructional model, Engage, Explore, Explain, Elaborate, and Evaluate, with engineering design as a sixth instructional component. Students first develop scientific understanding through investigation and then apply that knowledge to authentic engineering problems.

My proposed extension will investigate how AI can strengthen questioning, scientific investigation, conceptual understanding, engineering design, evidence-based reasoning, and formative assessment. It will focus on framework development, teacher professional development, and research-based evaluation and dissemination.

2. Foundation of the AI-IBED Model

Brand et al. (2017) developed IBED through professional development with Grades 6–12 mathematics, science, and career and technical education teachers. Participating teachers wanted engineering activities to advance scientific understanding rather than become engaging exercises disconnected from curriculum content. IBED therefore connects scientific inquiry with knowledge-based engineering solutions.

AI-IBED will retain that relationship. AI will serve as an instructional support tool, while scientific evidence, experimentation, conceptual reasoning, student agency, and teacher judgment remain central. The proposed extension has not yet been validated; its feasibility and educational effects will be investigated through the planned research.

3. Core Components of the Proposed Model

Curriculum content and learning objectives

Identify scientific concepts, engineering practices, learning goals, and assessment criteria. AI-assisted planning may help align standards, concepts, challenges, and assessment; teachers will verify accuracy.

Rationale and real-world relevance

Connect science content with authentic community problems. AI may help develop scenarios, locate accessible data, and adapt tasks to students' prior knowledge.

AI-supported scientific inquiry

Engage: contextualized phenomena and questions. Explore: experimental planning and data organization. Explain: guided questions and feedback. Elaborate: transfer to new contexts. Evaluation: formative feedback subject to teacher review. Students will verify AI-generated information against evidence.

AI-supported engineering design: the sixth E

Identify the problem and requirements; establish measurable criteria; brainstorm solutions; select concepts; prototype and test; justify a final solution; and reflect and improve. AI may support brainstorming, comparisons, modeling, and interpretation, but students must document and justify decisions with evidence.

Assessment and evaluation

Use conceptual assessments, design portfolios, student reflections, performance rubrics, and observations to evaluate science understanding, engineering reasoning, and critical AI literacy. Teachers retain responsibility for consequential assessment decisions.

Proposed instructional sequence: Curriculum objectives following the backward design principle → 5E scientific inquiry → Sixth E engineering design → Assessment, reflection, and redesign. AI-assisted support throughout the sequence, subject to teacher oversight. This sequence assists teachers to design a state aligned lesson plan and implement it in ways that help students to learn underlying scientific concepts of a problem, while moving ahead to design a prototype to address the problem. **This engagement seeks to bridge gap between knowledge acquisition and application.**

4. Teacher Professional Development and Implementation

I propose to develop professional development that prepares STEM teachers to design and implement AI-IBED lessons. Workshops will combine inquiry-based instruction, engineering-design planning, evaluation of AI-generated content, classroom implementation, and collaborative reflection. Teachers will learn to align activities with applicable curriculum standards and apply appropriate privacy, accessibility, and ethical safeguards.

The initial U.S.-based pilot will depend on securing school partnerships, resources, and necessary research approvals. Pilot findings will guide revisions to the model, instructional materials, professional development modules, and assessment instruments before broader implementation.

5. Research and Evaluation

The research will investigate how teachers integrate AI into inquiry and engineering design, how professional development influences instructional practice, how students demonstrate conceptual understanding and engineering reasoning, and which contextual factors support or constrain implementation.

Potential data sources include teacher interviews, lesson plans, classroom observations, student work, pre- and post-instruction assessments, and engineering-design portfolios. The design will use qualitative and quantitative methods appropriate to the pilot setting and sample. Implementation outcomes will be distinguished from learning outcomes: evidence that teachers use AI successfully will not by itself establish improvements in student achievement.

6. Dissemination and Broader Implementation

I intend to develop an implementation guide, adaptable lesson templates, professional development modules, classroom activities, assessment rubrics, and research publications. Appropriate open educational resources and revisions to the model may be shared through a public repository such as GitHub.

Dissemination will also be pursued through academic publications, professional conferences, educator workshops, and potential partnerships with universities and school districts. Materials will be designed for adaptation across science disciplines, grade levels, and technological contexts.